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RESEARCH ARTICLE

Charge Diagnostics and State Estimation of Battery Energy Storage Systems Through Transformer Models

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ABSTRACT With the continuous development of Artificial Intelligence (AI), designing accurate algorithms that provide diagnostics and maintenance of energy technologies is a challenging task in the energy transition domain. This research work focuses on the implementation of Transformer models for charge diagnostics and algorithm design of Battery Energy Storage Systems (BESSs). Experimentally, two Lithium-ion (Li-ion) battery cells were tested using a programmable DC electronic load to evaluate charge indicators, and 20 battery tests were performed for each cell. Filter, Wrapper, and Embedded methods are the techniques implemented to achieve Feature Selection and illustrate Key Performance Indicators (KPIs) in battery testing. Time series and state estimation are the Supervised Learning techniques executed for charge diagnostics and State of Charge (SOC) predictions. The results show remarkable performance metrics of the Transformer models, achieving over 94% accuracy in Model evaluation compared to traditional Deep Learning algorithms.

INDEX TERMS Deep learning, neural networks, battery energy storage system.

Term/Abbreviation	Definition
BESS	Battery Energy Storage System.
ECM	Equivalent Circuit Model.
AI	Artificial Intelligence.
NN	Neural Network.
Li-ion	Lithium-ion.
KPIs	Key Performance Indicators.
RS	Ohmic resistance.
R1	Polarization resistance.
C1	Polarization capacitance.
R2	Diffusion resistance.
C2	Diffusion capacitance.
VOC	Open circuit voltage.
Vcell	Voltage of the cell.

SOC	State of Charge.
RNN	Recurrent Neural Network.
MHSA	Multi-head self-attention.
LSTM	Long Short-Term Memory.
GRU	Gated Recurrent Unit.

I. INTRODUCTION

The energy transition is a promising topic that not only leads the scientific field towards a more sustainable future but also contributes to the emergence of new algorithms and strategies in the technological field. Electrification and smart applications constitute a beneficial renewable energy source for designing, developing, and implementing various types of energy technologies to replace fossil fuels with a Battery Energy Storage System (BESS), a core element in the operation of an electric vehicle.

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Different approaches have been considered to monitor the performance of a BESS, with Electrochemical Models, Equivalent Circuit Models (ECMs), and Mathematical Models being the most important to accurately predict Health and Charge indicators [1], [2], [3]. With the relentless strides in Artificial Intelligence (AI), the realm of informatics necessitates the meticulous development, rigorous verification, and seamless implementation of sophisticated algorithms to bolster the insights proffered by energy analysts across both industrial and academic fields.

In the Computer Science field, Neural Network (NN) architectures provide efficient and reliable solutions to predictive maintenance and the remaining useful life of a BESS; however, depending on the operation, battery properties, and experimental procedures, the NN's adaptability, robustness, and capability are challenging to achieve, which is why more advanced and improved architectures are currently being designed.

Several scientists and scholars have proposed Data-Driven approaches based on Machine Learning, Deep Learning, and Data Science methodologies [3], [4], [5], all to provide BESS modelling and diagnostics that support commercial consulting, research, and testing for enterprises. As highlighted by Gilbert Zequera [5], there exists a pressing imperative to delve into the design of algorithms capable of elucidating the behavior of a BESS by utilizing AI-powered technology, which introduces the research question to develop new AI models whose network mechanism not only make accurate predictions but also illuminate the intricacies of physical and virtual entities from an energy framework.

In the context of Deep Learning, renowned scientists have published scientific literature to highlight the relevance of AI methods and propose different categories of algorithms in the field of BESS. Time series models have also been the point of analysis for both evaluating and predicting the lifetime of Lithium-ion (Li-ion) batteries [6], [7], [8]. In 2020, Zhang et al. [9] proposed a Time series model for prognostics prediction based on a recurrent neural network, whereas, in 2021, Agab et al. [10] developed a unifying model to achieve forecasting by describing the dynamics of a BESS using learning Time series.

Due to different user needs and computational costs, efficient and robust algorithms have been developed in recent years to support the integration of Software Development and Computer Science in the engineering domain, one of them is an innovative Deep Learning model known as Transformer. A Transformer model is defined as a new network architecture whose operation works based on attention mechanisms. It was initially proposed by Vaswani et al. in 2017 [11] to achieve Natural language processing tasks, currently being able to support complex problems related to different AI topics, such as Computer Vision, Machine Translation, DNA sequence analysis, and Protein Structure analysis [11].

The implementation of a Transformer model in the BESS field has focused on energy consumption forecasting [12] charging demand prediction, and useful life evaluation [13],

[14]. However, the need arises to unify the energy domain and AI perspective in the algorithm design of a BESS, all to develop a methodology based on physical and virtual entities through various branches of knowledge such as Data Engineering, IT Operations, Electrical Engineering, and Software Development.

The main contribution to the scientific community is summarized in the following points:

- Complement the current Deep Learning algorithms associated with the prediction of Health and Charge indicators based on battery tests for Li-ion cells.
- Propose an innovative approach that considers an ECM, Transformer model, and Time series.
- Reduce computational complexity in algorithm design of a BESS due to AI methods.

Among the main reasons for developing a Transformer model are to improve the functioning of the network architecture through attention mechanisms, make the models more scalable and parallelizable in a BESS operation, and execute diagnoses based on experimental tests and Deep Learning approaches. The architecture and understanding of the Transformer model from a BESS perspective will be explored in the following sections.

In this article, two Li-ion cells were experimentally tested, and different datasets were collected by performing 20 battery tests for each cell, respectively, using a programmable DC electronic load. The motivation of this research is to develop an initial AI methodology using Transformer architectures that can provide modeling and diagnostics in battery management based on algorithm design.

The rest of the paper is organized as follows, in Section II, the problem statement, experimental battery tests, and motivation of this research through battery modelling are explained. Section III illustrates the calculation and identification of the Key Performance Indicators (KPIs) using different Feature Selection methods. Section IV implements the Transformer Model for state estimation and charge diagnostics of battery cells. Finally, in Section V a conclusion is presented to encourage the continuation of this research based on more advanced Transformer model architectures.

An overview of the proposed methodology is summarized in Fig. 1, which will be explained in detail in the following sections.

II. CASE STUDY: LEV50N BATTERY CELL TESTING

The datasets used in this article correspond to experimental tests of two Li-ion battery cells, initially measuring Voltage [V], Current [I], and time. The cells were tested using a standard constant current/constant voltage (CC/CV) charging protocol [15], [16]. During the charging process, a constant current rate of 0.80 C was applied until the voltage reached 4.10 V, at which point the voltage was maintained at 4.10 V until the charging current dropped below 0.50 A. For the discharging phase, the cells were discharged using a constant current method with a cut-off voltage of 2.75 V, ensuring

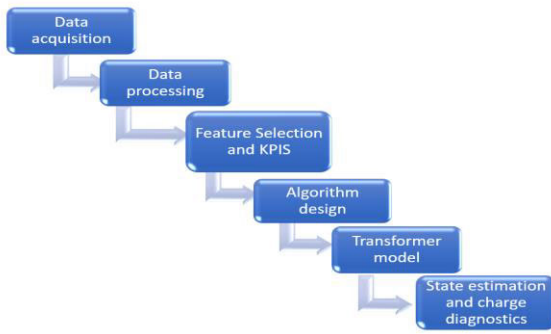


FIGURE 1. Flow-chart of the charge diagnostics and time series methodology.

a controlled discharge process. The total capacity of the cells was 50 Ah. All tests were conducted at room temperature (23 °C) under standard atmospheric pressure (1 atm), ensuring consistent test conditions to minimize any potential variations in performance.

The study employs an exhaustive Computer Science approach within the framework of TensorFlow and Keras as a high-level Application Programming Interface (API), experimentally subjecting the BESS to rigorous testing, complemented by meticulous performance data acquisition using a programmable DC Electronic Load.

The target application of this article is based on industrial, academic and research Deep Learning modeling complemented with static data, all to achieve the highest overall performance, optimal Training, and robustness of the algorithm.

A. BATTERY TESTS AND DATA ACQUISITION

A Mitsubishi i-MiEV battery pack, composed of 88 Li-ion cells type LEV50N and manufactured by GS Yuasa was collected from the innovative ISEAUTO project, an Estonian autonomous electric vehicle (AEV) deployed on the Tallinn University of Technology (TalTech) campus [17]. Specific information about the battery pack incorporates two modules that contain four cells, and ten modules integrated by eight cells, having a total amount of 88 Li-ion prismatic cells, which are connected serially using screwed contacts. Due to topics of strategic importance, energy, and cooperation, the battery pack was dismantled into several modules. A summary of the battery pack parameters and LEV50 cells is shown in Table 1 and Table 2.

TABLE 1. Battery pack specifications.

Manufacturer	GS Yuasa
Type	Li-ion
Number of cells	88
Pack weight	164.65 kg
System voltage	325.60 V

TABLE 2. LEV50N battery cell parameters.

Parameter	Value	Unit
Nominal voltage	3.75	V
Operating voltage range	2.75 to 4.1	V
1-hr rate typical Capacity 25°C	50	Ah
Charge voltage limit at 25°C	4.1	V
Charge termination threshold current	0.5	A

Different battery tests were performed experimentally to evaluate the performance of the cells using a programmable DC electronic load through the UltraLoad Software for remote operation and monitoring. The programmable DC electronic load has an adjustable current rising speed from 0.001 A/ μ s to 5 A/ μ s, a readback resolution of 0.1 mV and 0.1 mA, a list function that supports editing as many as 512 steps, and dynamic mode up to 30 kHz. A total of 20 battery tests for each LEV50N cell were carried out to evaluate the charge indicators of a BESS. Specifications of the battery tests consist of a slew rate of 0.001 A/ μ s, a step duration of 1 second, a frequency of 1 Hz, and a resolution of 0.8%.

B. SECOND-ORDER EQUIVALENT CIRCUIT MODEL

The ECM is a helpful approach that can simulate the operation of a BESS within an engineering framework. The framework is constructed by putting capacitors, resistors, and voltage sources in the circuit. In 2022, Gilbert Zequera et al. [1] conducted a literature review that explains the advantages and drawbacks of executing an ECM for electric vehicles and digital twin applications.

It is fundamental to mention that two different approaches based on an ECM can be considered to determine a BESS's dynamic behavior, including the voltage response to different load profiles and operating conditions [18]. The first-order ECM considers a capacitor in parallel, voltage source, ohm resistance, and polarization resistance. In contrast, the second-order ECM includes not only the same elements but also implements a polarization resistance, which is divided into electrochemical polarization resistance and diffusion resistance [18]. A graphical representation of a second-order ECM is illustrated in Fig 2.

The elements of a second-order ECM visualized in Fig. 2 are denoted by the ohmic resistance R_s , polarization resistance R_1 , polarization capacitance C_1 , diffusion resistance R_2 , diffusion capacitance C_2 , open circuit voltage VOC , and the voltage of the Li-ion cell V_{cell} . The mathematical derivation of the corresponding equations that explain the ECM from an engineering domain is found in [19] and [20].

In this research, a second-order ECM is coded using Python as the main programming language. Likewise, parameter estimation is achieved using the weighted mean of vectors algorithm [21], being a promising and innovative method that demonstrates superiority in terms of accuracy

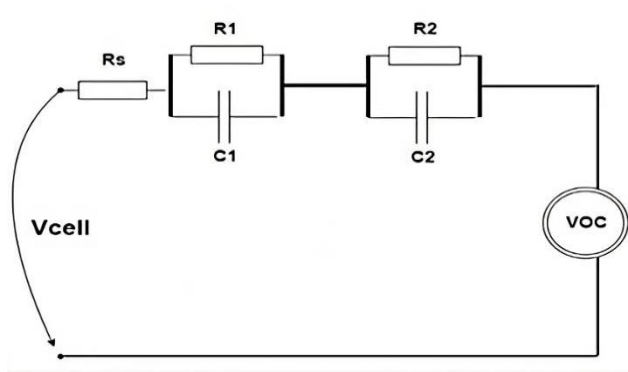


FIGURE 2. Graphical representation of a second-order ECM.

and speed in the performance of battery management systems.

To demonstrate the robustness of the weighted mean of vectors algorithm, Fig. 3 illustrates the predictions and compares them with experimental measurement data.

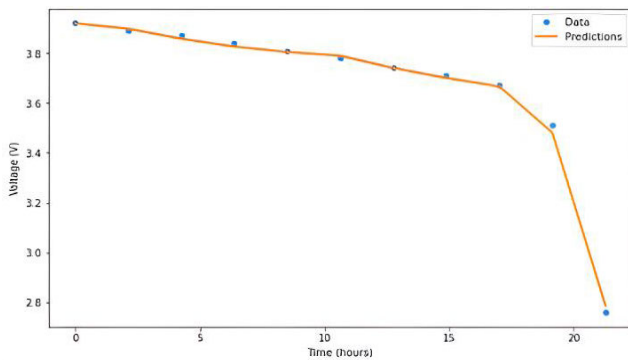


FIGURE 3. Graphical predictions of the weighted mean of vectors algorithm.

To validate the accuracy of the results through the weighted mean of vectors algorithm, performance metrics consisting of Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Mean Square Error (MSE) are calculated over the entire experimental measurements, which is provided in Table 3.

TABLE 3. Validation results of a second-order ECM.

Performance metrics	Results [%]
RMSE	0.0392
MAE	2.1510
MSE	1.9809

Regarding the simulation of the Li-ion cells, the State of Charge (SOC) is defined as the dependent variable and will be the main point of analysis for the following sections, also considered as the target variable in the predictions of the Transformer model. The second-order ECM is initialized to simulate the operation of a Li-ion cell, reflecting the evolution

of the SOC over time in hours. To provide a robust visualization and clear understanding of the simulation, the dataset has been labeled to classify the different states of a BESS. Fig. 4 shows the SOC representation based on a regression problem explained by Time series analysis.

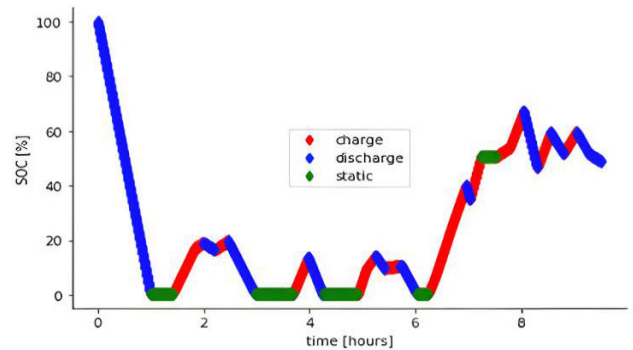


FIGURE 4. SOC evolution over time for a Li-ion cell using a second-order ECM.

III. KEY PERFORMANCE INDICATORS

Initially, the dataset is processed, and Feature Engineering is executed to obtain distinctive features intrinsically related to the battery tests, which are the Power [P], Resistance [R], Capacity [mAh], Energy [Wh], and SOC, the latter being considered as the predicted variable. The next steps consist of performing Exploratory Data Analysis to ensure that the dataset does not contain significant outliers, after that, different Feature Selection methods are executed to identify and calculate the KPIs.

Filter, Wrapped, and Embedded are the selected Feature Selection methods that will be implemented and discussed in this research. The objective of this step is not only to select the relevant features for model building, but also to understand the impact and mutual information that every feature shares during the different battery tests.

The KPIs obtained in the next sections refer to the features that best represent the relationship in the dataset, all to achieve the following tasks: 1) Reducing resource requirements and model complexity, 2) Reducing the size of the feature space, 3) Removing features that do not influence the predictions.

A. FILTER METHODS

Filter methods are a collection of techniques based on statistical approaches, whose main purpose is to measure the importance of every feature in the dataset. Fast processing and inexpensive computational cost are the main advantages of the Filter methods compared to Wrapper and Embedded, so when analyzing big data, they offer great benefits [22].

Among the most relevant categories of Filter methods are Correlation coefficients, Mutual information, Variance Threshold, and Chi-Square test. Because the features of the dataset are only numerical, without having any categorical

variable, Correlation coefficients, and Mutual information will be the Filter Methods implemented [22].

Correlation coefficients are Filter methods that help measure the linear relationship between two or more numerical features, whose main function is to display the correlation matrix between the features and the target variable [23]. Furthermore, correlation coefficients are beneficial in revealing information about collinearity, so it is strongly recommended to remove redundant features when they do not provide any additional information [23]. To select the most notable features using this method, a threshold is defined as a measure of the correlation value; when the correlation between the feature and the target variable is less than the threshold, the feature can be discarded.

A correlation map showing each Correlation coefficient for two different battery tests corresponding to cycles 1 and 20 respectively is illustrated in Fig. 5, showing that there is an elevated level of correlation between the independent variables and the SOC.

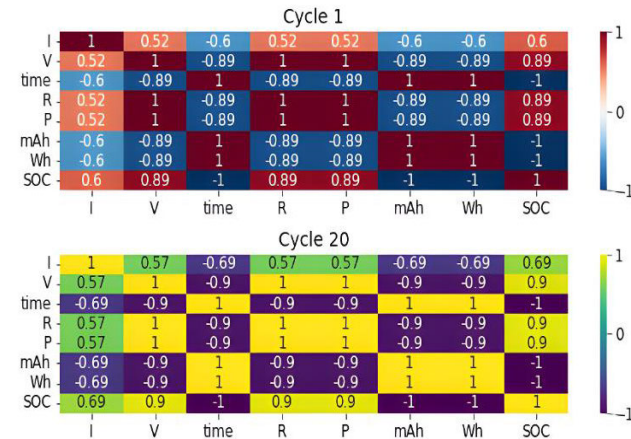


FIGURE 5. Correlation coefficients of two different battery tests. The top plot indicates cycle 1, and the bottom plot indicates cycle 20.

Considering the perspective of a BESS, battery tests are carried out to analyze the charging management of Li-ion cells during discharge, explaining this statistical behavior. Like Correlation coefficients, the Mutual information method is executed when the independent variables are numerical. It measures the level of dependence of the predictors on the target variable. A visualization of the Mutual information method is shown in Fig. 6, performed for cycles 1 and 20 of the battery tests, respectively. Notably, even though the cycles have changed, the Mutual information method shows that the scores share similar values for each corresponding feature in the dataset, demonstrating that there is an elevated level of dependence in terms of the Capacity and Energy of the BESS concerning the SOC.

The explanation for the different scores on Mutual information and Correlation coefficients is that correlation analysis provides a quantitative means to measure the strength of the relationship between the predictors and the target variable.

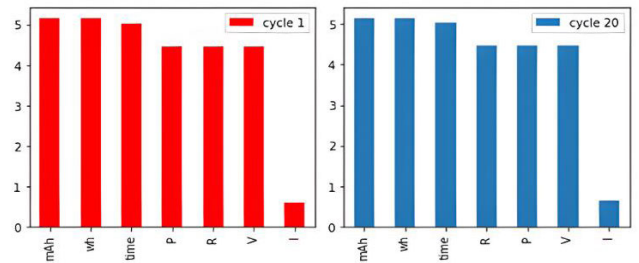


FIGURE 6. Mutual information of battery tests for cycles 1 and 20 respectively. The X-axis indicates the features; the Y-axis represents the score.

On the other hand, Mutual information is essentially the measure of the amount of knowledge that the user can obtain about a certain variable by knowing the value of another variable.

B. WRAPPER METHODS

Wrapper methods in Feature Selection provide an algorithm with the ability to evaluate the performance of a model on all subsets of features [24]. A significant advantage of Wrapper methods is the quality of learning with each combination of features, providing performance metrics for feature sets. However, selecting the most optimal features is a critical user task, all to avoid potential bias or vague physical interpretations of predictors during the Model evaluation step.

Forward Feature Selection, Backward Feature Elimination, and Exhaustive Feature Selection are the Wrapper methods implemented in this research. Forward Feature Selection operates by selecting the best variable among all the features and groups a different feature with the previously selected variable, iterating this process until the convergence criteria are completed, which usually specifies a quantity of features limit or final score on a performance metric. Backward Feature Elimination works on the opposite side of Forward Feature Selection, starting with all the features to evaluate the performance of a model and discarding irrelevant features that give a low level of performance, repeating the process until the convergence criteria are fulfilled. Exhaustive Feature Elimination works by searching for all combinations of features to evaluate the performance of a model over each subset of features, therefore, the combination of all features is the main result based on each corresponding performance metric [25].

Implementation of the Wrapper methods with their corresponding results are shown in Table 4. A minimum of three features with a maximum of seven features have been selected, and the metrics of performance is the RMSE for calculating the average score.

The results in Table 4 show that the maximum number of features gives poor performance on the Wrapper methods when considering Forward Feature Selection and Backward Feature Elimination. However, when the selected features are Voltage, Current, and time, there is a similar level of

TABLE 4. Wrapper methods results.

Feature subsets	Method	RMSE
Current, Voltage, time, Resistance, Power, Capacity, Energy	Forward Feature Selection	0.5054
	Backward Feature Elimination	0.4617
Current, Voltage, time, Resistance, Power, Capacity	Forward Feature Selection	0.2573
	Backward Feature Elimination	0.2131
Current, Voltage, time, Resistance, Power	Forward Feature Selection	0.2573
	Backward Feature Elimination	0.1764
Current, Voltage, time, Resistance	Forward Feature Selection	0.2145
	Backward Feature Elimination	0.1563
Current, Voltage, time	Forward Feature Selection	0.1719
	Backward Feature Elimination	0.1017
Voltage Resistance, Power	Exhaustive Feature Selection	0.1162

convergence in performance. A more detailed explanation of these preliminary results will be given at the end of this section.

C. EMBEDDED METHODS

Embedded methods, also known as Hybrid methods, are a category of Feature Selection algorithms that surpass the ability of Filter and Wrapper methods due to their efficient computational cost and reliability [24]. It is essential to specify that the process of searching for feature subsets in Embedded methods is incorporated as part of the Training step.

Embedded methods provide an operating mechanism between the Filter and the Wrapper because they combine the qualities of both approaches [25]. Like Filter methods, Embedded methods are computationally lighter than Wrapper methods, and all rely on the Training step where, during the classification or regression problem, the internal parameters are tuned to determine the appropriate weights given to each feature and produce the best accuracy [26].

Support Vector Machine (SVM), Artificial Neural Network (ANN), Decision Tree, and Random Forest are among the most relevant Embedded methods, which also consider the execution of regularization models, such as Lasso, Ridge, or Elastic net [26]. In this study, Gradient Boosted Tree and Random Forest are the two Embedded methods that will show the results of feature importance before starting the algorithm

design. The top three features with their corresponding scores for both the Gradient Boosted Tree and the Random Forest are visualized in Fig. 7.

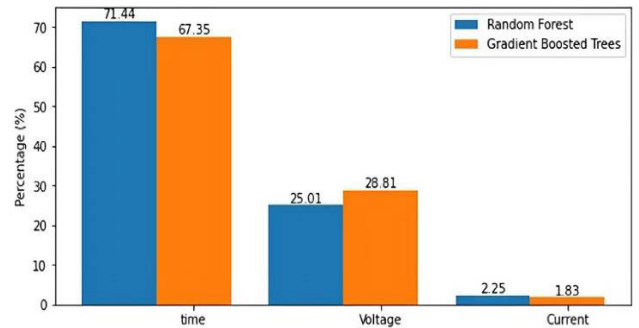


FIGURE 7. Graphical representation of the Embedded methods through Random Forest and Gradient Boosted trees.

D. SUMMARY OF INITIAL RESULTS

Regarding the results of the Filter methods, the Correlation coefficients show that Resistance and Power share an elevated level of correlation with the target variable, which is also in the same numerical range as Voltage, as do Capacity and Energy with time. Similarly, the Mutual information provides insight into the direct impact and intrinsic relationship between the Capacity and Energy consumption of the BESS during the SOC's evolution. Furthermore, based on the initial results and to reduce multicollinearity, only the most important features will be considered in the design of the algorithm, which is why it is required to execute the Wrapper and Embedded methods.

In the context of Wrapper methods, the larger the number of features, the lower the algorithm's performance compared to a minimum number of variables. In particular, the optimal number of feature subsets is achieved when the Wrapper method converges on three or four variables, so the previous results given by the Filter methods play a key role based on the level of dependence or independence of the predictors with the target variable. In addition, the highest performance is achieved by Backward Feature Elimination; therefore, the three selected features, which correspond to Voltage, Current, and time, will be the input variables in the network architecture. In case the user wants to select a larger number of features than recommended, the performance of the algorithm will decrease based on the Wrapper method results.

The Embedded methods demonstrate that the relevance of feature importance complements the results delivered by the Wrapper methods, allowing the size of the feature space to be reduced and features that do not influence the model predictions to be removed. Both Random Forest and Gradient Boosted trees show a level of convergence to identify and calculate the feature importance of the independent variables, not only based on the level of correlation the SOC but also the performance metrics of each feature subset. According to the results obtained by Embedded methods, Current, Voltage, and time are the most relevant predictors in the dataset and are

considered KPIs, all based on their property of storing and explaining the physical interpretation of additional variables such as Resistance, Power, Capacity and Energy of a BESS. These KPIs represent a remarkable level of statistical correlation in the set of predictors due to the Filter methods and show intrinsic engineering behavior with the target variable according to the Wrapper and Embedded methods.

It is required to mention that the results provided by the Feature Selection methods do not differ but rather have a different scope to complement each other in algorithm design, supported not only by statistical correlation and mutual information but also by the feature importance and performance metrics.

Before concluding this section, it is essential to highlight that KPIs will be defined as predictors during the design of the Transformer model in the next section, which will reduce computational complexity and promote a better understanding of the network architecture.

In the following sections, the proposed AI methodology in this research employs both virtual and physical entities. The physical entity refers to the Li-ion cells in the experimental tests, while the virtual entity is defined as the Transformer model in the algorithm design.

IV. TRANSFORMER MODEL

Recurrent Neural Networks (RNNs) have been one of the most prominent AI methods for capturing timely dependencies in sequences. As stated in [11], it has been scientifically and computationally proven that implementing only attention mechanisms without any RNN can improve the results of Deep Learning algorithms, all due to its ability to learn context and meanings through tracking relationships in sequential datasets.

Transformer models comprise large Encoder-Decoder blocks that use positional encoders to label data elements entering and leaving the network. Architecturally, attention units follow the data labels, calculating and mapping each related element. Attention queries are executed in parallel by calculating an equation matrix process, which is called multi-head attention.

From a BESS perspective, implementing a Transformer model by forgoing the decoder layer results in a more accurate state estimation [27], so this approach will be considered in the algorithm design that implements an Encoder-only architecture.

In this section, a Transformer model is designed, coded, and executed to achieve state estimation. The main objective is to show the robustness and adaptability of new NN architectures based on multi-head attention mechanisms and compare their performance with traditional Deep Learning algorithms, all to define a methodology for the algorithm design of a BESS through AI-powered technology.

A. ENCODER ARCHITECTURE AND ALGORITHM DESIGN

The encoder architecture is inspired in the Bidirectional Encoder Representation from Transformer (BERT) model,

in which attention layers process the input and provide a sequence of embeddings as output. Attention is a mechanism that assigns weights to each input based on their importance, where the weighted sum is then used to compute the output of the model [11]. In the NN design, the only difference with the Encoder-Decoder architecture is the self-attention: in BERT it is a bidirectional method that considers both at left and right with respect to the selected input.

The Encoder-only Transformer provides a faster Training step than RNNs, processing the information in common blocks, which are the input vector, positional encoding, encoder stack, and the linear layer. The input vector consists of Voltage, Current, and time, previously defined as KPIs. In contrast, positional encoding is a network representation that describes the position or location of the input in a sequence, such that each position is assigned a unique representation. The encoder stack is a network composed of multiple layers of autoencoders, and the linear layer indicates the matrix operations executed on the Multi-head self-attention (MHSA) mechanism [27], [28].

Mathematically, key, value, and query are the main elements of the MHSA [11], [28], [29], which are defined as K , V , and Q respectively. Dimension of the vector is defined as d_k and the variable T is represented as a set of building blocks that will serve as input to be mapped and processed in the model, usually named tokens in Natural Language Processing [29], [30]. The MHSA is explained by Equation (1).

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{Q * K^T}{\sqrt{d_k}}\right)V \quad (1)$$

In the context of the network architecture, the query refers to the projection vectors in the input sequence, the key represents every input associated with the possible predictions, and the values are the set of best predictions provided by the encoder.

After completing Feature Selection for KPIs identification, the data is processed, and the next steps summarize the algorithm design of the Transformer model within the framework of TensorFlow and Keras as a high-level API:

- *Data processing:* A class containing a set of objects is defined in Python to process the data, based on the properties of the BESS, such as the rated capacity and the vector array of the KPIs. Regarding the hyperparameters of the Transformer model, the initial network architecture is defined as including a set of hyperparameters, such as batch size, learning rate, epochs, weight decay, and attention units.
- *Encoder layer:* The encoder layer is defined using the fully connected sequential layers, composed of a MHSA mechanism, and followed by a fully connected feed-forward network; furthermore, batch normalization is added to significantly improve the efficiency of the multivariate Time series process [29].
- *Transformer:* The Transformer model is designed by using the full encoder, subsequently the network starts the execution of the embedding layer by passing the

input and using the positional encoding method. A learning rate schedule with Cosine Annealing is developed through Adam optimizer [31] for monitoring the losses between the start and end of every epoch.

- **Training:** The corresponding hyperparameters are incorporated into the Transformer model to start the Training step. The algorithm compiles the values defined in the Data processing, updates, and saves the best model parameters, obtaining the Training and Validation curves.
- **Fine-tuning and Hyperparameter optimization:** The initial set of hyperparameters is processed to start the Fine-tuning process, and hyperparameter optimization is executed through the Keras tuner. In this step, the performance metrics per every combination of hyperparameters are calculated, and the resulting network architectures correspond to the minimal validation loss of the learning curves.
- **Validation and Testing:** K-fold cross-validation is performed, and minimum validation loss is achieved, after that, the performance metrics are obtained. Finally, new predictions are calculated on unseen datasets in the Model evaluation step.

An overview of the algorithm design and the previous explanations can be seen in Fig. 8.

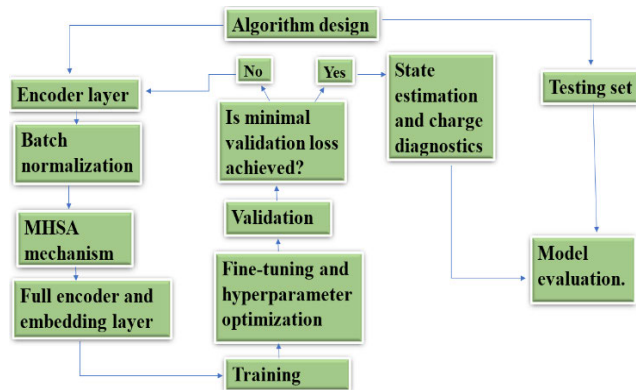


FIGURE 8. Flowchart of the algorithm design of the Transformer model.

B. TRAINING AND FINE-TUNING

In the design of the algorithm, the Training set helps the model to recognize patterns and make predictions, on the other hand, the Validation set serves as a reference point to evaluate the performance of the model based on the learning process. In this study case, the Training set corresponds to the battery tests of the first Li-ion cell in a total of fifteen cycles, while the Validation set comprises the remaining five cycles. In the context of network architecture, Transfer Learning plays a crucial role that leverages the patterns, representations, and knowledge learned by an existing model on a large and often generic dataset, therefore saving time and computational resources, improving generalization, and allowing for learning with limited data. Noteworthy, a visualization

learning curves is a critical task during the Fine-tuning, so that ensures the model is enabling rapid development and iteration to avoid overfitting or underfitting.

An example of learning curves of the Training and Validation losses is seen in Fig. 9, indicating that the model reaches a minimum loss and convergence above 100 epochs.

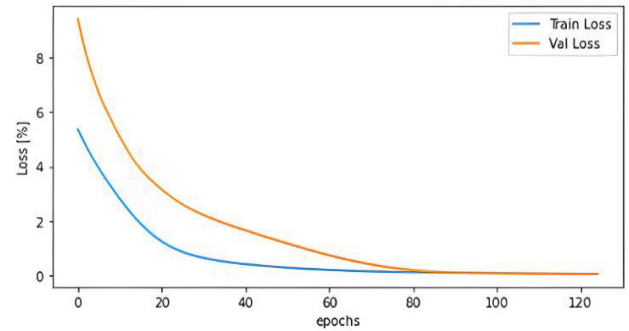


FIGURE 9. Learning curves of the Training and Validation losses.

The Transformer is executed to make predictions and test performance with other RNNs and Bidirectional Recurrent Neural Networks (BRNNs). For comparison purposes, Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU) [5], [32], [33], and their bidirectional architectures BiGRU and BiLSTM [34] are implemented to evaluate the accuracy of different Deep Learning algorithms. Performance metrics are obtained by calculating the MSE, RMSE, and MAE. Table 5 shows the performance metrics in the Validation step.

TABLE 5. Performance metrics in the validation step.

Algorithm	MSE [%]	RMSE [%]	MAE [%]
Transformer	0.0100	1.0031	0.8554
LSTM	0.0334	1.8422	0.9529
GRU	0.1100	3.3166	2.1503
BiGRU	0.0676	2.6012	1.7629
BiLSTM	0.01450	1.2062	0.7077

Upon the successful completion of the Fine-tuning and Hyperparameter optimization processes, the optimized network architectures are meticulously saved and stored in Keras file formats. These refined models are then ready to be utilized during the subsequent Model Evaluation phase. In this critical step, the models are applied to the Testing set, allowing for an in-depth assessment of their generalization ability. Table 6 provides the optimal network hyperparameters for each algorithm.

C. STATE ESTIMATION AND CHARGE DIAGNOSTICS

To complete the Model evaluation step, the Testing set consists of the twenty battery test cycles of the second Li-ion cell.

The Transformer model is evaluated on the Time series dataset generated by the second-order ECM and explained

TABLE 6. Optimal network hyperparameters.

Algorithm	Bath size	Learning rate	Epochs	Weight decay	Units
Transformer	32	0.0010	100	1e-4	8
LSTM	64	0.0008	51	1e-5	150
GRU	64	0.0007	65	1e-5	250
BiGRU	64	0.0004	42	1e-5	200
BiLSTM	64	0.0060	75	1e-5	250

in Section II, whose operating parameters were previously processed by the battery tests of the Li-ion cells.

The metrics of performance are calculated for RNNs, BRNNs, and Transformer in the Testing sets to complete the Model evaluation step. MSE, RMSE, and MAE are calculated to obtain the results for each algorithm and are provided in Table 7.

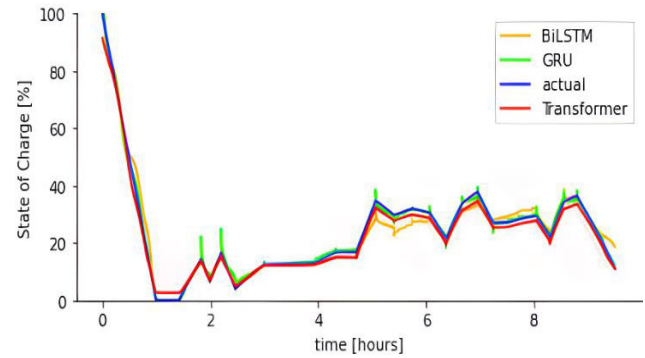
TABLE 7. Performance metrics in the model evaluation.

Algorithm	MSE [%]	RMSE [%]	MAE [%]
Transformer	0.3318	5.7072	3.3460
LSTM	1.7810	11.5134	9.2970
GRU	2.0482	12.5514	10.5919
BiGRU	1.9169	11.7673	9.8786
BiLSTM	1.5521	10.6758	8.8536

Fig. 10 shows a Time series representation illustrating the evolution of the SOC, offering a comprehensive visual analysis of the performance of the Transformer and the most accurate NNs in comparison to the actual values observed in the testing set. This representation stands as a powerful tool for the real-time monitoring of a BESS, providing essential diagnostics for effective charging management. By identifying anomalous patterns, it enables early detection of potential issues within the SOC, thereby enhancing the reliability and efficiency of the system. Moreover, this capability facilitates predictive maintenance, optimizing the lifespan and performance of the BESS while minimizing downtime and operational risks.

The error rate of Transformer is significantly lower than that of RNNs and BRNNs with less than 6% in terms of RMSE and MAE, and 1% for MSE, which represents a crucial result of the Transformer architecture, specifically considering the parallelization and Training efficiency due to MHSA by excluding the information loss during positional encoding.

Compared to RNNs and BRNNs, Transformer provides scalability for large datasets and prominent global pattern recognition, implying that, unlike traditional NN architec-


FIGURE 10. Graphical representations of the predictions in the model evaluation step.

tures, it efficiently processes data in parallel and captures relationships between all elements of a sequence simultaneously, due to the advantage of using MHSA. Consequently, Transformers can better model complex relationships between distant tokens in the input generated by the positional encoding. In terms of efficiency in Training and Validation steps, Transformers are more efficient than RNNs and BRNNs in handling long-range dependencies in sequences, which offers promising results for studying and designing different architectures such as Decoder-only and Encoder-Decoder architectures.

Regarding BESS charge diagnostics, RNNs process charging and discharging data one element at a time, maintaining an internal hidden state updated at each step. BRNNs operate recursively, where the output at each step depends on the previous hidden state and the Voltage and Current values stored in the query. On the other hand, Transformers non-sequentially process the charging management data in parallel, relying on self-attention mechanisms to capture time dependencies between different elements in the Voltage and Current vectors, without recurring connections or hidden states.

D. DISCUSSION

In the initial steps, Feature Engineering and Feature Selection play the most significant role, all to provide information about the correlation, dependence, and mutual information of the variables before algorithm design. It was demonstrated through Filter, Wrapper, and Embedded methods that even if an elevated level of correlation is found, the algorithm shows the maximum level of performance using Voltage, Current, and time, achieving KPI calculation.

From the BESS domain, including all predictors and target variables allow the user to understand the operating profile and physical interpretation, however, not in all cases these indicators are available, which is why the KPI calculation and Feature Selection are required to ensure data processing and gain insight into algorithm design. Considering the architecture of the Transformer, Encoder-only is selected for effective state estimation, mutual information between

hyperparameters and the MHSA elements was established based on the positional encoder, in which the learning rate schedule and the batch size monitor the behavior of the SOC due to different step sizes and samples that update the model weights in the MHSA mechanism. Positional embeddings are considered another innovation introduced in a Transformer model to replace recurrence, the idea is to use fixed or learned weights that encode information related to a specific position of a token.

In the Model evaluation step, the Transformer surpasses both RNNs and BRNNs in performance metrics. Its algorithmic design, leveraging the MHSA mechanism, excels at learning complex, non-linear patterns during BESS operation, delivering exceptional performance in model-dependent state estimation algorithms. This sophisticated mechanism proves advantageous for tackling more demanding tasks such as forecasting. However, to unlock its full potential, modifications to the network architecture are essential, particularly through the implementation of encoder and embedding layer structures or Fine-tuning techniques.

Although the Encoder-only Transformer shows outstanding performance metrics by executing Fine-tuning and Hyperparameter optimization, implementing additional search algorithms is an opportunity area to complement the network architecture due to the execution of more techniques such as Bayesian optimization, genetic algorithms, or Hyperband.

Compared to existing Machine Learning models and physics-based approaches [35], [36], this methodology starts with algorithm design from a beginner level that initially familiarizes the reader with the topic, until reaching an MHSA mechanism that can make accurate predictions in the Model evaluation step. Furthermore, considering the algorithm design, due to the high level of network architectures and programming tools of this research, this methodology complements some Deep Learning methodologies and Transformer models that not only focus on the Remaining Useful of a BESS [37], [38] but also the State of Health [39], fault prognosis [40], and charging management [41].

The challenges addressed by the current methodology in the predictive maintenance of a BESS are comprehensively summarized in Table 8. The innovative approach leverages advanced algorithm design, incorporating predictive models capable of analyzing vast amounts of operational data in real time. By utilizing sophisticated techniques in AI, the Transformer not only anticipates diagnostics before they escalate by Transfer Learning but also adapts to evolving patterns within the BESS. This dynamic framework ensures that the predictive maintenance strategy is both proactive and responsive, thereby significantly reducing downtime and extending the operational lifespan.

The Encoder-only Transformer represents a crucial and initial step in the future algorithm design of a BESS using AI methods, all to execute optimal predictive maintenance and support business consulting, research, and enterprise testing.

TABLE 8. Challenges addressed by the current methodology.

Topic	Challenges addressed
Battery technology	- Extensive battery testing of Li-ion cells validated in an R&D environment - Virtual entity composed of an Encoder-only Transformer, whose physical representation is based on battery properties
Algorithm design of a BESS	- Accurate Model evaluation based on MHSA mechanism and KPIs of a BESS - Innovative design using Time series analysis, operating data, and AI methods.

The promising results and contributions of this research demonstrate the potential for the build, design, and operation of a BESS over its lifetime, supporting not only the global demand for transportation electrification [42] but also the possibility of employing 80% of their initial capacity for multiple storage solutions and second use, such as high-energy, high-power applications, and hybrid solutions [43].

Employing reliable, adaptable, and accurate AI methods in the digitalization of the energy industry is a driver for both industry and academia [44], so considering the second use of a BESS is a challenge that enhances climate change mitigation and promotes future opportunities on the path to the energy transition [45].

V. CONCLUSION

In this study, a Transformer model based on an Encoder-only architecture was employed to achieve state estimation and charge diagnostics, marking a crucial step in the algorithm design of a BESS. Extensive testing of Li-ion cells was carried out under various conditions and scenarios, accompanied by measurements via UltraLoad software for remote operation and monitoring. The simulation was run by coding a second-order ECM to generate Time series data and test the robustness of the algorithms. This strategy was carried out through the design, development, and implementation of the MHSA mechanism to convert the initial networks into several AI models, capable of emulating the operation of a BESS.

Key results demonstrate that the design and execution of a Transformer model shows more optimal performance compared to RNNs and BRNNs in the Deep Learning domain, whose network architecture can improve the algorithm design within an energy framework and explain the behavior of both virtual and physical entities of a BESS, answering the research question. Compared with current scientific references, this article proposes an innovative and exceptional approach in the field of a BESS by using an Encoder-only Transformer, which is complemented not only by Feature Selection methods and Time series analysis but also by an

engineering perspective based on a second-order ECM and experimental battery tests of Li-ion cells.

The resulting Encoder-only architecture, firmly rooted in empirical data, is a significant outcome of this research that represents a substantial advance in the development of AI-driven technology for BESS diagnostics. Regarding the limitations of this work, the BESS chemistry is an important element to consider in the physical entity, and the Encoder-only architecture in the algorithm design, so it is recommended to experimentally test new chemistries and design different Transformer architectures in future work.

Among the most important novelties of this research are: 1) The proposed Transformer can be used as part of advanced model-dependent state estimation algorithms of a BESS through AI methods, 2) As the algorithm design of the Transformer has physical and virtual entities, the virtual portion can be optimized by Black-box AI, while the physical entity can be used for simulating different experimental conditions, 3) For current analysis of the BESS behavior, the Time series representation helps in the anomaly detection and charge diagnostics, and is also complemented by prior performance and comparison of different Feature Selection methods to identify KPIs, 4) The compatibility of the Encoder-only Transformer through the MHSA mechanism makes it feasible to exploit different architectures such as Decoder-only and Encoder-Decoder.

The pioneering and visionary research carried out in this article represents a testimony to the unification of several branches of knowledge, such as Electrical Power Engineering and Mechatronics, Computer Science, and DevOps. The next steps will consist of the deployment through Machine Learning Operations (MLOps) and the incorporating of new Transformer architectures.

Finally, it is of utmost importance to mention that the resources of a research group must facilitate the participation of those resources for other research groups. Our research must facilitate a modest contribution to promote ties of cooperation and collaboration, which is our vocation as scientists, researchers, and scholars.

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